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STEREOTACTIC FRACTIONATED RADIOTHERAPY IN THE TREATMENT OF JUXTAPAPILLARY CHOROIDAL MELANOMA: THE MCGILL UNIVERSITY EXPERIENCE

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Purpose: To report our experience with linear accelerator-based stereotactic fractionated radiotherapy in the treatment of juxtapapillary choroidal melanoma.
Methods and Materials: We performed a retrospective review of 50 consecutive patients diagnosed with juxtapapillary choroidal melanoma and treated with linear accelerator-based stereotactic fractionated radiotherapy between April 2003 and December 2009. Patients with small to medium sized lesions (Collaborative Ocular Melanoma Study classification) located within 2 mm of the optic disc were included. The prescribed radiation dose was 60 Gy in 10 fractions. The primary endpoints included local control, enucleation-free survival, and complication rates.
Results: The median follow-up was 29 months (range, 1–77 months). There were 31 males and 29 females, with a median age of 69 years (range, 30–92 years). Eighty-four percent of the patients had medium sized lesions, and 16% of patients had small sized lesions. There were four cases of local progression (8%) and three enucleations (6%). Actuarial local control rates at 2 and 5 years were 93% and 86%, respectively. Actuarial enucleation-free survival rates at 2 and 5 years were 94% and 84%, respectively. Actuarial complication rates at 2 and 5 years were 33% and 88%, respectively, for radiation-induced retinopathy; 9.3% and 46.9%, respectively, for dry eye; 12% and 53%, respectively, for cataract; 30% and 90%, respectively, for visual loss [Snellen acuity (decimal equivalent), <0.1]; 11% and 54%, respectively, for optic neuropathy; and 18% and 38%, respectively, for neovascular glaucoma.
Conclusions: Linear accelerator-based stereotactic fractionated radiotherapy using 60 Gy in 10 fractions is safe and has an acceptable toxicity profile. It has been shown to be an effective noninvasive treatment for juxtapapillary choroidal melanomas. **© 2011 Elsevier Inc.**

Choroidal melanoma, Juxtapapillary, Local tumor control, Stereotactic radiotherapy, Toxicity.

INTRODUCTION

Choroidal melanomas arise from melanocytes in the choroid and account for 85% to 90% of all uveal melanomas. Although rare, with an annual incidence of 6 to 7 cases per 1 million people, the disease is nevertheless the most common primary intraocular malignancy in the adult population (1, 2).

Standard treatment previously consisted of complete removal of the tumor by enucleation. However, studies have failed to demonstrate any advantage of enucleation over eye-conserving therapy in reducing metastasis or in improving overall survival (3, 4). The management of choroidal melanomas is aimed at tumor control with organ

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